

```

1 Program Question_4
2   Implicit none
3   Integer::a,b,result
4   Open(Unit=15,file='a1q4_in.txt')
5   Open(Unit=17,file='a1q4_out.txt')
6
7   Read(15,*)a,b
8   if(a<=0.Or.b<=0)then
9       Write(17,*)"Invalid: Please Enter the positive Integr value"
10      else
11          result=gcd_calculate(a,b)
12      Write(17,*)"The gcd of",a,"and",b,"is",result
13
14  end if
15
16
17
18  contains
19  Recursive function gcd_calculate(a,b) result(gcd)
20      Implicit none
21      Integer, INTENT(IN)::a,b
22      Integer::gcd
23      If(Mod(a,b)==0)then
24          gcd=b
25      else
26          gcd=gcd_calculate(b,mod(a,b))
27
28      End If
29  end function
30
31 End Program

```